Classification Evaluation Metrics

Accuracy, Precision, Recall and F1-Score

1. Why Do We Need Evaluation Metrics?

After building a classification model, we need to check how well the model is performing.

Suppose a model predicts whether a patient has a disease.

The model may make four types of predictions:

- Correctly predicts disease.
- Correctly predicts no disease.
- Predicts disease when the patient does not have it.
- Predicts no disease when the patient actually has it.

Therefore, simply counting the number of correct predictions may not always be enough.

Important idea:
Different applications have different costs for wrong predictions.
Therefore, we choose an evaluation metric according to the problem scenario.

2. Positive and Negative Classes

For a binary classification problem, we normally have two classes.

Term	Meaning
Positive	The class that we are interested in detecting.
Negative	The other class.

Example: Disease Prediction

- Positive = Disease
- Negative = No Disease

The meaning of positive and negative depends on the problem.

For example, in spam detection:

- Positive = Spam
- Negative = Not Spam

3. Confusion Matrix

A confusion matrix compares the actual class with the predicted class.

	Predicted Positive	**Predicted Negative**
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

3.1 True Positive (TP)

The actual class is positive and the model also predicts positive.

$$\boxed{\text{Actual Positive} \rightarrow \text{Predicted Positive}}$$

Example:

A patient has the disease and the model predicts "Disease".

3.2 True Negative (TN)

The actual class is negative and the model also predicts negative.

$$\boxed{\text{Actual Negative} \rightarrow \text{Predicted Negative}}$$

Example:

A patient does not have the disease and the model predicts "No Disease".

3.3 False Positive (FP)

The actual class is negative, but the model predicts positive.

$$\boxed{\text{Actual Negative} \rightarrow \text{Predicted Positive}}$$

Example:

A healthy patient is predicted to have the disease.

This is also called a **false alarm**.

3.4 False Negative (FN)

The actual class is positive, but the model predicts negative.

$$\boxed{\text{Actual Positive} \rightarrow \text{Predicted Negative}}$$

Example:

A patient has the disease, but the model predicts "No Disease".

This can be particularly serious in disease screening.

4. Example: Disease Prediction

Suppose a model is tested on 100 patients.

The confusion matrix is:

	Predicted Disease	**Predicted No Disease**
Actual Disease	15	5
Actual No Disease	10	70

Therefore,

$$TP = 15, \qquad FN = 5, \qquad FP = 10, \qquad TN = 70$$

Total patients:

$$15 + 5 + 10 + 70 = 100$$

5. Accuracy

Accuracy tells us the proportion of all predictions that are correct.

$$\boxed{Accuracy = \frac{TP + TN}{TP + TN + FP + FN}}$$

For the disease example:

$$Accuracy = \frac{15 + 70}{15 + 5 + 10 + 70}$$

$$Accuracy = \frac{85}{100} = 0.85$$

$$\boxed{Accuracy = 85\%}$$

What does Accuracy ask?

"Out of all predictions, how many were correct?"

When is Accuracy useful?

Accuracy is useful when:

- Both classes are reasonably balanced.
- False positives and false negatives have similar importance.
- We want an overall measure of correctness.

Example:

Suppose a quality-control system classifies manufactured products as:

- Good
- Defective

If both types of errors have approximately similar consequences and the classes are reasonably balanced, accuracy can be a useful metric.

6. Precision

Precision focuses on the predictions made as positive.

$$\boxed{Precision = \frac{TP}{TP + FP}}$$

For the disease example:

$$Precision = \frac{15}{15 + 10}$$

$$Precision = \frac{15}{25} = 0.60$$

$$\boxed{Precision = 60\%}$$

What does Precision ask?

"When the model predicts Positive, how often is it actually Positive?"

Precision is affected by **False Positives**.

If FP is high, precision decreases.

Example: Spam Detection

Suppose an email classifier predicts that 100 emails are spam.

If many genuine emails are incorrectly classified as spam, important emails may be sent to the spam folder.

Here, false positives are undesirable.

Therefore, we care about:

$$\boxed{\text{High Precision}}$$

The idea is:

"When I say Spam, I want to be reasonably sure that it is actually Spam."

7. Recall

Recall focuses on the actual positive cases.

$$\boxed{Recall = \frac{TP}{TP + FN}}$$

For the disease example:

$$Recall = \frac{15}{15 + 5}$$

$$Recall = \frac{15}{20} = 0.75$$

$$\boxed{Recall = 75\%}$$

What does Recall ask?

"Out of all actual Positive cases, how many did the model find?"

Recall is affected by **False Negatives**.

If FN is high, recall decreases.

Example: Disease Screening

Suppose a model is used to identify patients who may have a serious disease.

Missing a patient who actually has the disease can be dangerous.

Therefore, we want to detect as many actual disease cases as possible.

Here, we care about:

$$\boxed{\text{High Recall}}$$

The idea is:

"I do not want to miss actual positive cases."

8. Precision vs Recall

Precision and recall focus on different types of errors.

Metric	Focus	Main Concern
Precision	Predicted Positive cases	False Positives
Recall	Actual Positive cases	False Negatives

A useful way to remember them:

Precision:
"Of the cases I predicted as Positive, how many were really Positive?"
Recall:
"Of the cases that were actually Positive, how many did I find?"

9. F1-Score

Sometimes both Precision and Recall are important.

In such situations, we can use the F1-Score.

$$\boxed{F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}}$$

For the disease example:

$$Precision = 0.60$$

$$Recall = 0.75$$

Therefore,

$$F1 = 2 \times \frac{0.60 \times 0.75}{0.60 + 0.75}$$

$$F1 = \frac{0.90}{1.35}$$

$$F1 = 0.6667$$

$$\boxed{F1 \approx 66.67\%}$$

What does F1-Score tell us?

F1-Score provides a single measure that considers both **Precision and Recall**.

A high F1-Score generally requires both precision and recall to be reasonably high.

Example: Fraud Detection

Consider a system that detects fraudulent transactions.

Two types of mistakes are important:

- **False Positive:** A genuine transaction is incorrectly marked as fraud.
- **False Negative:** A fraudulent transaction is not detected.

Both types of errors matter.

Therefore, looking at only precision or only recall may not give the complete picture.

F1-Score can be useful when we want to consider both.

$$\boxed{\text{Fraud Detection} \rightarrow \text{Precision and Recall both matter}}$$

10. Why Accuracy Can Be Misleading

Consider a disease dataset containing 100 patients:

- 99 patients do not have the disease.
- 1 patient has the disease.

Suppose the model predicts "No Disease" for every patient.

The results are:

	Predicted Disease	**Predicted No Disease**
Actual Disease	0	1
Actual No Disease	0	99

Therefore,

$$TP = 0, \qquad FN = 1, \qquad FP = 0, \qquad TN = 99$$

Accuracy:

$$Accuracy = \frac{0 + 99}{100} = 99\%$$

So the model has:

$$\boxed{Accuracy = 99\%}$$

This looks very high.

But recall is:

$$Recall = \frac{0}{0 + 1} = 0$$

$$\boxed{Recall = 0\%}$$

The model failed to detect the only patient who actually had the disease.

Important:
A very high accuracy does not always mean that a classification model is useful. This is especially important when the classes are highly imbalanced.

11. Which Metric Should We Use?

The choice depends on what type of error is more important in the application.

Metric	Use When	Typical Scenario
Accuracy	Overall correctness is important	Balanced classification / quality classification
Precision	False Positives are costly	Spam detection
Recall	False Negatives are costly	Disease screening
F1-Score	Both FP and FN matter	Fraud detection

12. Simple Memory Trick

Metric	Remember It As
Accuracy	Overall correctness
Precision	"Predicted Positive → How many were actually Positive?"
Recall	"Actual Positive → How many did we find?"
F1-Score	Balance between Precision and Recall

13. Formulas at a Glance

Metric	Formula
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$
Precision	$\frac{TP}{TP+FP}$
Recall	$\frac{TP}{TP+FN}$
F1-Score	$2\frac{Precision \times Recall}{Precision + Recall}$

14. Final Understanding

When evaluating a classification model, do not immediately ask:

"What is the accuracy?"

First ask:

"What type of mistake is more important in this problem?"

Then select the metric accordingly.

Overall idea

- **Accuracy** → Overall correctness
- **Precision** → Avoid unnecessary Positive predictions
- **Recall** → Avoid missing actual Positive cases
- **F1-Score** → Consider both Precision and Recall